

STUDENT PROJECT ASSIGNMENT

Exploratory Data Analysis on a Cars Dataset

Clean, explore, visualise, and explain meaningful patterns in automobile data.

Project Type	Dataset	Submission
Individual project	Mentor-provided or any suitable cars dataset	1 code file + 1 PDF

Project Goal: Perform a complete Exploratory Data Analysis to understand the dataset, improve data quality, identify patterns and outliers, and communicate useful insights through clear visualisations. A machine-learning model is not required.

☐ Project Requirements

Dataset Choice

- Use the dataset provided by the mentor or select any suitable public cars dataset.
- The dataset should contain useful automobile attributes such as brand, year, price, fuel type, mileage, engine size, transmission, or horsepower.
- Mention the dataset source clearly in the PDF report.
- Choose a dataset large enough to support meaningful analysis.

Required EDA Workflow

1. Inspect the shape, columns, sample rows, and data types.
2. Check missing values, duplicates, and inconsistent categories.
3. Clean the dataset and explain each important decision.
4. Calculate descriptive statistics for numerical and categorical columns.
5. Perform univariate, bivariate, and multivariate analysis.
6. Identify outliers, relationships, and at least five useful insights.



☐ Questions Students May Explore

- Which brands or fuel types are most common?
- How are car prices distributed?
- How does vehicle age relate to price?
- Does mileage appear to influence price?
- Do automatic and manual cars differ in price?
- Which features are strongly correlated?
- Which brands have the highest average price?
- Where are the important outliers or unusual records?

Note: These questions are examples. Students should select questions that suit their dataset and avoid forcing an analysis when the required columns are unavailable.

☐ Minimum Analysis Requirements

Data Understanding and Cleaning

- Show dataset dimensions, column names, and data types.
- Separate numerical and categorical variables.
- Check duplicate rows and missing values.
- Correct unsuitable data types and inconsistent labels.
- Treat missing values or outliers only when justified.
- Record the dataset size before and after cleaning.

Visualisation and Insights

- Include a minimum of six meaningful charts.
- Use at least four chart types, such as histogram, bar/count plot, box plot, scatter plot, or heatmap.
- Every chart must have a clear title and readable labels.
- Write a short observation below every chart.
- Present at least five final findings supported by the analysis.
- Do not make claims that the dataset cannot support.

Evidence expected in the PDF: Dataset overview, data-quality checks, before/after cleaning information, descriptive statistics, charts, written observations, key insights, limitations, and conclusion. Do not paste the full code into the report.



☐ Submission Requirements

Submit exactly **two files**:

- **File 1:** One code file (.py or .ipynb)
- **File 2:** One PDF project report

The assignment document contains no code. Students must write, run, and understand their own analysis.

File 1 - Code File

1. Submit exactly one code file in Python or Jupyter Notebook format.
2. Keep all data loading, cleaning, analysis, visualisation, and findings in the same file.
3. Use clear section headings, meaningful variable names, and short comments.
4. Recommended filename: StudentName_Cars_EDA_Project.py or StudentName_Cars_EDA_Project.ipynb.
5. The file must run successfully using the submitted or clearly referenced dataset.
6. Remove unrelated practice code and unnecessary output before submission.

Important Submission Rules

- Submit original work and be ready to explain the complete workflow.
- Dataset sources, references, and borrowed ideas must be acknowledged.
- Do not copy and submit a complete EDA project without understanding it.
- Do not submit extra code files or the dataset unless the mentor requests it.
- Screenshots and charts in the PDF must be readable.



□ File 2 - PDF Project Report

Recommended length: 5 to 8 pages. Include charts, tables, screenshots, observations, and explanations. Do not paste the full code into the report.

Section	What to Include / Cover
1. Cover Page	Project title, student name, batch, date, and dataset source.
2. Objective	Purpose of the EDA and the main questions selected for analysis.
3. Dataset Overview	Rows, columns, feature descriptions, data types, and sample records.
4. Data Quality and Cleaning	Missing values, duplicates, outliers, type corrections, and cleaning decisions.
5. Descriptive Analysis	Summary statistics and important observations for numerical and categorical variables.
6. Visual EDA	Minimum six charts with titles, labels, and a written observation for each chart.
7. Key Findings	At least five evidence-based insights, plus limitations of the dataset.
8. Conclusion	Overall learning, challenges faced, and possible future analysis.



□ Evaluation Rubric

Area	Marks	Evaluation Focus
Dataset understanding	15	Correct interpretation of columns, data types, and dataset context.
Data cleaning	20	Accurate checks, appropriate treatment, and clear explanation of decisions.
EDA and statistics	20	Relevant descriptive and relationship analysis.
Visualisations	25	Meaningful charts, variety, readability, and observations.
Insights and conclusion	10	Evidence-based findings, limitations, and clear conclusion.
Code quality	10	Organisation, naming, readability, and successful execution.
Total	100	



☐ Final Submission Checklist

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| ☐ Dataset source is mentioned | ☐ At least five final insights are presented |
| ☐ Missing values and duplicates are checked | ☐ Only one code file is submitted |
| ☐ Minimum six charts are included | ☐ PDF contains explanations, not full code |
| ☐ Every chart has an observation | ☐ Student can explain the complete project |

Final Note: Late, incomplete, copied, or non-working submissions may receive reduced marks.

Submission Date: _____

Batch: _____

Cars EDA Student Project | Prepared by Sruthi Tarimana |



RIVOQUIX LEARNING PRIVATE LIMITED
Hustlehub Tech Park,
Somasundarapalya Main Rd, adjacent
27th Main Road, Sector 2 hsr layout,
Karnataka 560102



+91 91 083 21780
info_hr@unloxacademy.com



www.unlox.com
CIN U85500KA2025PTC204246



@UnloxAcademy